

# Johny Yong Jun Siang

[Johnnyong10@gmail.com](mailto:Johnnyong10@gmail.com) | +65 93726829 | <https://johnnyong.github.io/DeveloperPortfolio/> | [www.linkedin.com/in/johnnyong](https://www.linkedin.com/in/johnnyong)

## SUMMARY

---

Software Developer and B.Sc. (Hons) Interactive Media & Game Development student with over **5 years** of experience in **C++, C#, Python (intermediate)**, and **JavaScript**. Strong background in **game systems engineering, multiplayer networking**, and **custom engine development**, with hands-on experience in web development using **React, HTML**, and **CSS**. Extensive experience with **Unity 3D**, including **VR development, gameplay systems, shader programming**, and cross-platform deployment. Proven ability to take ownership of both team-based and solo projects across the full development lifecycle, from system design to deployment.

## NOTABLE PROJECTS

---

### UrbanRun (Temporary)

- Collaborative project with a 5-member team, developed in Unity 3D (2026) to showcase at SIT's Open House. Designed and implemented the game's **systems architecture, state management, and core gameplay mechanics**.
- Inspired by *Subway Surfers*, focused on fast-paced endless runner design tailored for event demonstration.
- As of now this project remains **work in progress**, serving as a platform to stay current with Unity 3D 2026 features and practices.

### Grabity

- Led a 9-member team to develop a physics-driven top-down survival game built on a **custom C++ engine**, managing the product roadmap and technical milestones.
- Designed the engine's **Entity Component System (ECS)**, including a Game Object Factory and a Tag/Layer Manager to optimize object handling and performance.
- Authored comprehensive **Technical Design Documentation** using **MkDocs**, maintaining a live version via GitHub Pages for team alignment and version control.

### Obelisk

- Led a large-scale interdisciplinary collaboration **over 3 months** in **Unity 3D**, originally targeting PlayStation 4 VR before pivoting to **Oculus Quest** due to COVID-19 restrictions. Directed technical pipelines and coordinated cross-discipline efforts with artists to ensure cohesive delivery.
- Implemented Oculus VR integration, designed grid-based interaction systems, developed VR input handling, and created custom shaders.
- Navigated scope changes and production challenges, successfully delivering a complete VR game within a tight timeline.
- Key learning experience in adapting workflows under real-world constraints and leading a mixed-discipline team to completion.

## EDUCATION

---

**B.Sc.(Hons), Interactive Media & Game Development**  
Singapore Institute of Technology / DigiPen Institute of Technology

**Diploma in Game Development and Technology**  
Nanyang Polytechnic

~August 2023 - April 2027

~April 2018 - April 2021

## SKILLS

---

**Programming & Scripting:** C, C++, C#, Python (Intermediate), Java (Intermediate), HTML, Lua, JavaScript.

**Networking & Backend:** Photon Networking, Forge Networking, Firebase, Azure DevOps, JSON

**Software & Tools:** Visual Studio 2022, Visual Studio Code, Git, GitHub, Android Studio, Flutter, MkDocs, React, [Node.js](#), Unity 3D, Unity

**Project Management & Design:** Technical Design Documentation, Cross-functional Team Leadership, UI/UX Design, Agile Workflow

**Spoken Languages:** English (Fluent), Mandarin (Fluent)

## ACHIEVEMENTS

---

### Edusave Good Progress Award

Nanyang Polytechnic

### Commanding Officer's Coin

Singapore Armed Forces

## EXPERIENCE

---

### Mentor, Singapore Institute of Technology Community Challenge

Singapore Institute of Technology - Part Time

~October 2025 - October 2025

- I volunteered to mentor teams in using **VittaScience AI** posture & image recognition tools for prototype development.
- Supported the students across 5 days on their programming workflows, feature planning, UI/UX design decisions, and technical troubleshooting.

### Auto Technician

Singapore Armed Forces - National Service

~May 2021 - August 2023

- Maintained and repaired military vehicle systems as an auto technician
- Collaborated within a technical team under strict deadlines and safety protocols
- Developed a storage management mobile app during National Service using Azure DevOps & Flutter for development and Firebase for real-time tracking of resource usage.

### Software Engineer

YJP Surveyors Pte Ltd - Internship

~June 2020 - August 2020

- Contributed to a HoloLens mixed reality project built in Unity, enhancing visualisation of underground pipes and connections normally hidden by walls or flooring.
- Implemented **voice recognition** features to improve user interaction and accessibility.
- Gained hands-on experience with point cloud data and **LiDar technology** to support spatial mapping and alignment.
- Delivered a fully functional prototype that was commended for its effectiveness and innovation.

## CERTIFICATIONS

---

### Playstation Game Development Training Completion

- Successfully completed industry recognised training in Playstation workflows by utilising the Playstation DevKit.

### AI For Industry: Literacy in AI

- Gained foundational knowledge of artificial intelligence concepts and their industry applications.